Sales Forecasting & Demand Prediction Dashboard Report

Executive Summary

This report presents a complete Sales Forecasting & Demand Prediction Dashboard developed using Power BI and Excel. The project focuses on forecasting future sales trends, predicting customer demand, optimizing inventory planning, and improving strategic business decisions using historical sales data.

Project Objective

The objective of this project is to build a professional forecasting dashboard that helps organizations estimate future demand, improve inventory planning, and reduce business uncertainty using predictive analytics.

Industry Relevance

Forecasting dashboards are widely used in retail, FMCG, manufacturing, and e-commerce industries. Companies like Walmart, Amazon, Nestlé, and ITC rely on demand forecasting for supply chain optimization and production planning.

Business Problems Solved

• Prevent stockouts and overstocking
• Improve sales forecasting accuracy
• Optimize procurement and inventory
• Identify seasonal demand spikes
• Support revenue planning and business growth

Tools & Technologies Used

• Power BI
• Microsoft Excel
• Power Query
• DAX Functions
• Forecasting Techniques
• Data Visualization

Dataset Overview

The dataset contains historical sales transactions with fields such as Order Date, Product Category, Units Sold, Sales Amount, Marketing Spend, Forecasted Sales, Actual Sales, Stock Availability, and Demand Level.

Data Cleaning Process

The following data cleaning activities were performed:
• Missing value treatment

• Duplicate removal
• Date formatting
• Product category standardization
• Outlier detection
• Validation checks
• Month, Quarter, and Year extraction

Key Performance Indicators (KPIs)

• Total Sales
• Total Units Sold
• Forecasted Sales
• Actual Sales
• Forecast Error %
• Demand Growth %
• Sales Growth %
• Stock Availability

Dashboard Visualizations

• KPI Cards
• Sales Trend Line Chart
• Forecast vs Actual Sales Comparison
• Product Category Demand Chart
• Region-wise Demand Analysis
• Seasonal Sales Pattern Analysis
• Forecast Error Analysis
• Stock Risk Indicators

Business Insights

• High-demand products identified
• Seasonal sales peaks detected
• Inventory shortage risks highlighted
• Forecasting accuracy measured
• Demand planning gaps analyzed

Professional Dashboard Layout

Top Section: KPI Cards displaying Total Sales, Forecasted Sales, Actual Sales, Forecast Error %, and Demand Growth.
Middle Section: Forecast vs Actual Sales Charts, Product Demand Analysis, and Regional Insights.
Bottom Section: Stock Risk Analysis, Seasonal Trend Charts, and Forecast Accuracy Insights.
Side Panel: Filters and slicers for Region, Product Category, Year, and Season.

Resume-Ready Project Description

Developed a Sales Forecasting & Demand Prediction Dashboard using Power BI to analyze historical sales data, forecast future demand, monitor inventory risks, and improve strategic business planning.

GitHub Upload Files

• Dataset (.csv / .xlsx)
• Power BI Dashboard File (.pbix)
• Dashboard Screenshots
• README.md
• Final Project Report

LinkedIn Showcase

Share dashboard screenshots, business insights, forecasting visuals, and GitHub repository links to showcase analytics and business intelligence skills professionally.

Dashboard Components & Purpose

Dashboard Component	Business Purpose
KPI Cards	Quick executive summary of sales performance
Forecast vs Actual Chart	Measure forecast accuracy
Demand Analysis	Identify high-demand products
Regional Insights	Analyze region-wise sales trends
Stock Risk Analysis	Prevent stockout and overstock issues
Seasonal Trends	Detect peak demand periods

Conclusion

This project demonstrates practical knowledge of forecasting analytics, demand prediction, business intelligence reporting, and inventory optimization using Power BI. It is highly relevant for aspiring Data Analysts, Business Analysts, Sales Analysts, and Supply Chain Analysts.